

Technology Domain 3

based on use Case
⊕ Budget

pricing options

on-demand .1

down to the second

use Cases

app under development

unpredictable workload

no upfront or long commitment

can't be interrupted

workload will not run more than year

Spot .2

unused EC2 Capacity

use Cases

your app fees at very low price

start/stop time

[can be interrupted]

on-demand 90% على السعة
السر يتغير كل ساعة

Reserved Instances .3

Commit to specific instance type for 1 or 3 years

use Cases

your app needs capacity reservation

upfront discount

steady state usage

Compute #

rent & manage Virtual servers

EC2

Elastic Compute Power

note

Instances not considered serverless

EC2 Instance يقوم بتسليم

preconfigured template (Amazon Machine Image [AMI])

Free tier 750 Compute hours free

use Case

Deploy DB and have full control .1

Deploy web app .2

Access methods to EC2

AWS Management Console .1

Secure shell (SSH) .2

most common way via key pair

EC2 Instance Connect (EIC) .3

AWS System Manager .4

via browser based shell or CLI

← **Commitment** : **72%** **مقبوض** ← **Compute Usage** غير على ال Usage

← **Capacity Reservation** وفي الحالة دي مقبوض كنتي [و دا منطقك بتجوز نوع استخدام اوي كان مستقل على اوي]

Features

← بيكون معاها

Elastic Load Balancing

← **distribute traffic between multiple instances** ودا ادهو

← عندي منه انواع

← **Application load Balancer (HTTP/HTTPS) layer 7**

← **Static DNS, Routing HTTP Gateway load balancer layer 3**

← **Route 53 Firewall**

← **Classic Load Balancer (layer 4 & 7)**

← **Network load balancer (High Performance for TCP/UDP) layer 4**

← **static DNS application static IP** لقيت اوي اوي مشي static DNS application static IP

EC2 Auto Scaling

← **instances automatically based on Demand** ← **replace** ← **add** ← **instances**

← **across AZs**

← **Horizontal Scaling (scale out)** ← **instances** ← **Horizontal Scaling (scale out)** ← **instances** ← **Horizontal Scaling (scale out)**

← **Vertical scaling (scale up)** ← **instances** ← **Vertical scaling (scale up)** ← **instances** ← **Vertical scaling (scale up)**

Reserved Instances

← **75%** ← **مقبوض**

← **AZ** ← **مقبوض**

← **All upfront** ← **مقبوض**

← **partial upfront, no upfront**

← **Convertible** ← **option** ← **types**

← **4** ← **مقبوض**

Dedicated Host

← **Physical server** ← **مقبوض**

use cases

← **Compliance** ← **مقبوض** ← **multi-tenancy model**

← **server-bound licence from vendor like microsoft**

← **[pre-socket, per core, per-VM]**

← **70%** ← **مقبوض** ← **VMs** ← **run** ← **as a service**

Dedicated Instances

Saving Plans

← **Compute usage per hour for 1 or 3 years**

use cases

← **Flexibility to change instance type, OS, ...** ← **مقبوض**

← **multiple compute services**

Lambda execute Code in response to event .3

Lambda Function has 15 minutes timeout .4

Serverless Compute ← **Fargate** .3
engine For Containers manage Containers like Docker

Same app
Placed in Container → بحسب الك ما
Services deployed على Transport
تأثيره لا يتغير شيئاً زى ما هو مكتوب
Functionality

Serverless Fargate زى Lambda هي برودو
Scales automatically

Quickly launch resources for small projects ← **LightSail** .4

Deploy سهل
preConfigure apps
Word Press زى

simple screen for people with no cloud experience
provides low predictable monthly fee
\$3.50

similar to Service
AWS Quick Starts

Serverless Compute ← **Lambda** .2
Service to write Code without Manage Servers
Lambda scales automatically

application ال
Code للخط جوا
Functions
building block for many
Serverless app
Process Server

Realtime File Processing .1
زى ما بترفع Files على S3
وتقوم برفع Lambda
trigger لتفعل Action معين

sending email notifications .2

Backend business logic .3

alexq زى القائد جوا

Features
1. اللغات اللى بتمسكها Lambda هي
Java, Go, Powershell, Node.js,
C#, Python, Ruby

2. تقدر تكتب ال Code فى ال environment
تأخذك locally وتعمله upload او
تكتبه جوا ال Lambda مباشرة
Console

these services provide a place to store data that accumulated over years for analysing

← **Storage #**
S3
simple storage service

S3 object storage service that is highly available

objects (Files) are stored in Buckets (Directories)

S3 is unlimited storage can hold millions of objects

objects can be private or public

تقدر تـ upload objects من خلال

CLI أو Console
programmatically using SDKs

security can be set at bucket level or individual object using level

ACLs bucket policies

access point policies
أو تطبيق أو سمع بعضا عادي

enable Versioning
هذه feature في S3 وهي

to create multiple versions of your object [to protect against accidental deletion]

⊕ to [use previous version]

allow you to run cloud services in your DC (on-premises)

← **out posts** .5

النوع
من data
need on premises due to latency (or)

data sovereignty needs
[data residency] اهـ

هو بـ **hybrid experience**
dilemma install
ال Servers عندك في DC
بتاعتك و تستخدم اوى
في ال

بتتقارن مع ال services
ال على ال cloud عنان تطبيقا على
ال servers بتاعتك
Through APIs
to develop apps on-premises

← **Batch** .6
هو يستخدم مع
ال workload الكبير

بعمله processing عن

طريقا ان ال workload الكبير دا مقسم

small chunks or batches
و بتستعمل بالتوالي

مع بعضا عنان السند بـ العمل اسرع

→ Run hundreds or thousands of small batch processing jobs

→ Dynamically provisions instances based on volume

You can also setup **Cross-region Replication** setup ولكن لازم انت انت الـ AWS by default
 دا بنفسك و AWS by default
 برا الـ Region الـ انت اختار بقا في الاول إلا
 لو انت القوت.

S3 storage classes ← مختار based على
 Budget الـ use Case

- General purposed storage recommended for [Frequently accessed data]
 - Data stored accross [multiple AZs]
 - offers low latency + high throughput
- "انتاجية عالية" ودا لانه عند الـ رقم الـ standard
 في الـ durability [11 9s] والـ availability [99.99%]

2. S3 Intelligent Tiering ← use ML behind the scene
 automatic moves data to most cost-effective storage class. [automatic Cost Saving]

← يكون استخدم النوع الـ recommended وى
 مع الـ [Data with unknown or changing access patterns] **few app**
 ← ملحوظة مهمة جدا وانه النوع دا عليه
No retrieval fees → ودا لانه بيهدى الـ
 حاجة automatic فالتالى
 no need for user to manually manage retrieval
 وطلبنا تحديث ليهل كذا فعملنا فى الـ cost
 - Data stored accross multiple AZs
 with 11 9s durability & 99.9% availability

S3 ←

You can use **access logs** to [track the access of buckets & objects]

S3 is **Regional Service**, but bucket names globally unique.

Durability ← يعني الـ data موجودة في مكانها
 ومضاعش ومضاعش حاجة
never lost or compromised
 كـ "يتم اختراقها وسرقها
 والمساومة عليها"

S3 Designed For (11 9s of durability) → 99.999999999%

Availability ← يعني اتاحه الـ data وقت ما اتجى اعمل
 access عليها
 (important to access data quickly when you need it)

S3 Designed For (99.99% availability)

← طبها ازاى عندها durability وavailability عالية كذا
 [Data stored inside bucket in region will be replicated accross multiple servers in the same region] طبها

Glacier Instant Retrieval .5

For long-term data storage and archival for lower cost (long-term backups لا سي)

Data retrieval with milliseconds when needed

Data stored across multiple AZs with durability 11 9's
ومعاني حاجة من
المتاح لا سي (لا سي) access
كل فينا و فينا

Minimum storage duration [90 days] عندنا

Glacier Flexible Retrieval .6

long-term archived لا يرد ولا
backups باعرجها سي

Data retrieval [3 options] ليها

- 1-5 minutes → expedited
- 3-5 hours → standard
- 5-12 hours → bulk (is free)

Data stored across multiple AZs with the same durability and availability نفس الموضوع

minimum storage duration 90 days يردو

Glacier Instant Retrieval
data retrieval
في نفس ال
بين العزق في وقت ال

S3 standard - Infrequent Access (IA) .3

For less frequent data access but [requires rapid access]

milliseconds access when needed

Cheaper than standard option هو

و ببسخدم ال [long-lived Data]

Data stored across multiple AZs with 11 9's durability & 99.9 availability.

S3 one zone - Infrequent Access (IA) .4

For less frequent data access but rapid access when needed (in milliseconds) يردو

minimum storage duration 90 days عندنا
Standard IA

20% cheaper than standard-IA option هو

[stored in one AZ] لكن هتبقى في

و اقل ال availability 99.5%
دقيقت 12 بين باعرجها لوال AZ
طارت فالdata طارت معانا.

option هو ال مناسب ال
- Re-Creatable data
- Accessibility and durability not essential

← تابع S3 outposts :

لـ عشان بقى عزلة في الـ on-prem

فبعت الـ data ← stored across multiple devices and servers

لـ ومفيدة durability و availability مطلقاً

← النوع recommended اوى في الـ استخدام

مع الـ applications الـ هي [Demanding performance needs and you need to keep data close to your on-premises apps]

← بتتطلب اراء عالمي ومحتاجة البانا جنبها

← S3 use cases

1. ← S3 for static web sites
⊕ use CloudFront for global distribution

2. Data archive using Glacier

3. store data for analytics with Athena & Redshift

4. ← With mobile application apps to upload files (to speed this process use S3 Transfer Acceleration)

⌘ لاحظ انك تقدر تتفاد بين كلاس S3 Classes using S3 LifeCycle.

7. S3 Glacier Deep Archive

← برود الـ long-term data ,

archival accesses

لـ بزيادة لـ هنا (accessed once or twice a year)

← بيكون اكثر بيس الـ data الـ انت

[Retaining data for compliance requirements]

← طبعا دا لـ خيلى [the cheapest option of all S3 options]

Data Retrieval has [2 options]

1. 12 hours

« نهار يوم »

2. 48 hours

« يومين »

← برود Data accessed multiple AZs with the same durability and availability

← عندنا [Minimum Storage duration 180 days]

8. S3 outpost : provide object storage on premises for data that needs to be kept local.

← وطبعا عشان لها شفالة on-premises

ملوش نفس الـ Capabilities لو انا شفالة على الكلاود ، مفندي

فقط الـ Standard Storage Class

store long-term.
production
data which
need quick accessible

Use Cases for EBS

[most Common storage service for EC2]

Running a data base on Instance
(وفق الحالة في حالة Volume EBS)

Instance store

is a local store
that physically
attached to
the host computer

ذاكرة
ال EBS
في الميزينج
instance
واحدة

[يعني هو ال VM ال instance
ال قابلية لتقالبي على نفس
ال storage physical server
that physically
attached to
the instance

ميزته انه سريع جدا

[The fastest option for data access
and make higher I/O speeds]

لكن فيه عيب خطير جدا وهو ان

التخزين فيه temporary

[اول ما يتوقف ال instance
data lost]

Use Cases for Instance store

Temporary
storage needs

[Data replicated across instances
multiple instances
تتبع البيانات
بيو data replication]

[بني chatgpt يقول ان المعلومة
دي فلت delete]

Storage Services

EC2 storage

all instances must have a
root drive [وذا بيغ يكون
حاجة من اثنين
EBS أو Instance
store]

اما لو انت عايزه حاجة
shared that associated with many
instances وفق الحالة دي استخرم

EFS

فنتكلم عن كل حاجة بالتفصيل بقي

Elastic Block Store

is a device called (volume)
that can attach to instance

[attach only على بالظ انه بيد
one instance with in the same
AZ معاه]

سبب الفكرة ال يعني ممكن ال
الواحدة يبقى عندها اكثر من volume

Data فيه يتكون persist

(تأبقة مبيتطيرش حتى وانت عامل stop
لل instance)

[Tied to one AZ هو
من region
حتى]

فانت لو عاوز ال data تبقى موجودة
على اكثر من AZ you have to replicate
the block storage

is hybrid storage ← **Storage Gateway** 5

Service when you want to store some data on the cloud, and keep some data on-premises with low latency [Connect on-premises data with cloud data "using site-to-site vpn direct connect" على حسب انت واصل مع ال cloud ازاي]

use Cases

1. Moving backups to cloud
2. Low Cost with hybrid storage
3. Need to access data with low latency.

helps to manage ← **AWS Backup** 6

data backups across services like EC2, EBS, EFS وغيرهم
[مفني بتساعدك انك تـ create backup plan لـ services وال resources دي
لك data ال عليها يعني وفي ال plan لتجدر:]

- ① How Frequently you would backup it (Frequency)
- ② How long would you retain the data (Retention)

Elastic File System: EFS 4

is a serverless network File system for sharing files (حاجة كدا عالمة زي Google drive وكتاب access على)

instances support ال support ال linux file system only

accessible from different AZs ولكن in the same Region

[تيد كون ال EBS كان 1 AZ]

more expensive than EBS

له هو اغلى من ال EBS عشان كدا مش هتستخدمه غير في الصنعة بي

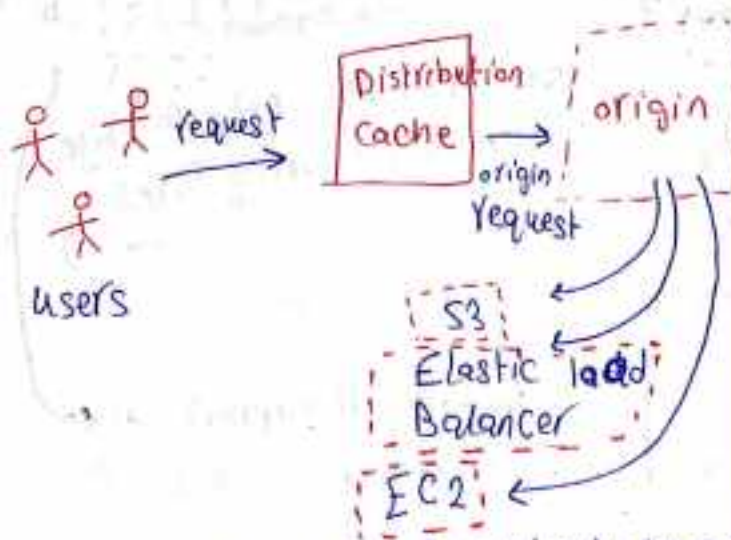
EFS use Cases

1. Main directories for business critical apps (critical apps ال امصاحبة shared directories)

2. Lift and shift existing enterprise apps

(يعني هتستخدم ال EFS لما تبجي نقل migration existing من ال on-premises app تنقله على ال cloud بدون اى تعديلات في الكود تقريباً ودا معنى (lift and shift)

كدا ال فوات كان ال storage service ال EC2 عدا ال S3 هتبدأ تفع في حاجات تانية في ال storage



← يقول هنا انه لما بيجي request من ال user
 يروح يدور على ال Content في ال
 Distribution Cache
 الاول، ولو مش موجود في ال
 request من ال Distribution Cache
 للمور العلي الداتا (حايه منه)
 ال هوميكن بيقي EC2, S3, ELB
 ويبيها caching الاول وبعد ما يرجع
 ال response ال user.

Use Cases

1. S3 static website, deploy the Content globally
2. prevent attacks, like DDOS
3. IP address blocking [CloudFront can prevent users in certain countries from accessing the Content]

Content Delivery

Content delivery network (CDN)

is a mechanism to deliver Content quickly and efficiently based on your geographic location

← with low latency
 وقت كبير ما بين request وال respond
 ← request على ال services بتاعنا بيقي ؟

is a CDN that CloudFront delivers data and applications with low latency.

make Content available globally or restricts it based on location

ال Content داممكن بيقي static or dynamic web Content (mainly for static Content)

use edge locations to Cache Content [means copy of a file]

→ edge location like a mini DC where files are cached

→ [Collection of edge locations named "Distribution Cache"]

← عنوان سريع delivery هي اصله يستخدم
[Globally distributed edge locations]
← لو انت عندك اكثر من Customer تقدر تخلط
around the world

يعملو upload لا Central bucket
الانت عليها
عادي جدا.

Networking *

Virtual Private Cloud ← VPC

allow to create secure
private network in aws cloud where
you launch your resources.

← فانت بيها تقدر تقا
Private Virtual Network

وتقدر ان IP وال Subnets وال security
Groups ⊕ Configure route tables
وهكذا...

← انت بت launch فيها او VPC هتق
isolation your resources
and protection
لا resources

← لاحظ ان ال VPC بتاعتك هتد على اكثر
من AZ جوا نفس ال Region عادي
[VPC spans AZs in a Region]

← عندي ال VPC جواها Sub Components:
ب قولنا اننا بتد على اكثر من AZ وتعرف
ال AZ عامة هي: AZ Consists of one
or more physical separated DC
with Connectivity housed in
separate facilities.

2. Amazon Global Accelerator

mainly used for applications
that use EC2 or Net Workload
Balancer

[يعني من الاجز لتسرع وصول
ال users لا dynamic يتعلق
app
ال انت صايف في مكان معين]

→ send your users through
AWS global network (Lags)
when accessing your content
and speed up delivery as
[it routes the traffic to
the nearest healthy endpoint
based on health checks ⊕
Routing Policies]

Route ال traffic
جوا ال AWS global network
على طول ال Region او المكان ال انت حاط
فيه ال Content بشكل اسرع، وحسنت
ال Performance بنسبة 60%

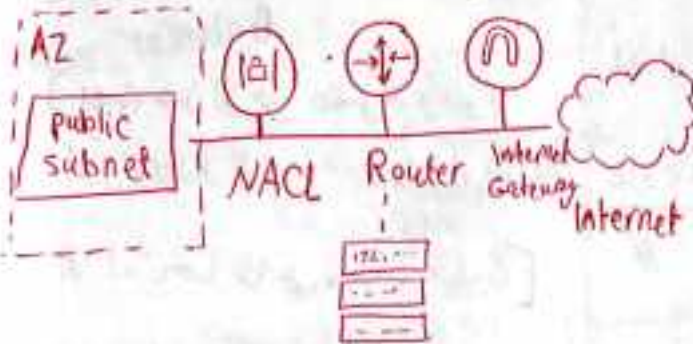
(كافيه Router كبير وبتدقش بي host
هو ابي حاجة)

3. S3 Transfer Acceleration

→ improves content upload
and download from S3 buckets.
[fast transfer files over
long distances]

Internet Gateway

allow public traffic to the internet from the VPC



وهو داسي ادا
نقشه منو انا
و عاوزو
Route
اروح لينا

Route Table

subnets

attached

attached route

IGW & table

VPC Peering

Peering allow to connect 2 VPCs

together via AWS Private network

to make them behave [يهم واحد] as 1



از VPC Peering لوانا بوسه ما بين 2 و 2 على
از cloud و اما لوعاوزو اعلى Hybrid
model و on-premise على طريقين

VPC SubComponents etc

Subnet

allow you split the network inside the VPC, subnet واز

في المكان الجواب باظبط
[launch your resources]

private subnets

الresources ال
المشاعوز يكون عليها access
من ال internet و ال Backend
Data base و

public subnets

هي داي accessible

From the internet

اصلا بتبقى public ازاي

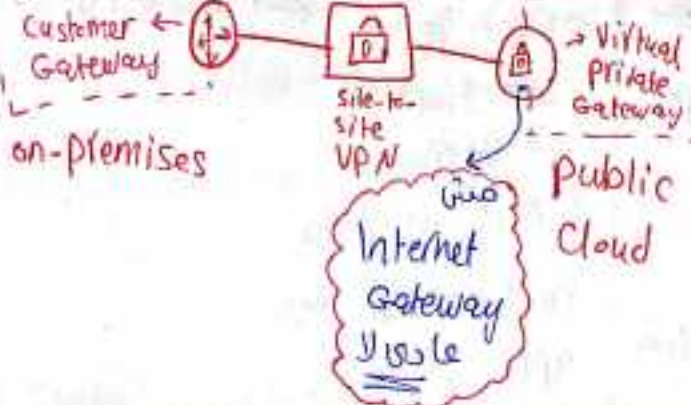
بستخدم another component
يخلوان في access من برا مسموح
على ال subnet دي زي:

Network ACL (NACL)

لهدا ال هيفلتر ال traffic
ال داخل على ال subnet دي
ويشاء كد ان ال traffic كند

Router & Route Table

that defined where the network traffic is directed



AWS Direct Connect
 ← ودا بيكون
 Physical توصيل
 مش من على
 الانترنت

[is a dedicated physical network connection from your on-premises DC to AWS]

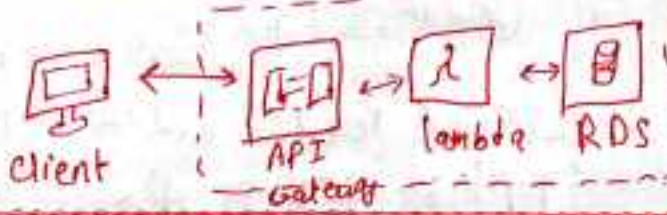
تاليه Networking Services بعد ال vpc وحواراتها

Route 53
 ← ال route اعوزني نكلم عن ال DNS
 [DNS directs the Internet traffic by connecting domain names with web servers]
 ← هي عبارة عن DNS service that route users to the application

Domain name registration
 ← تقدر كمان نقل من على
 يعني بختج اسم ال Server ال بي run عليه
 ال app تابعك بدل مات ال route عليه IP

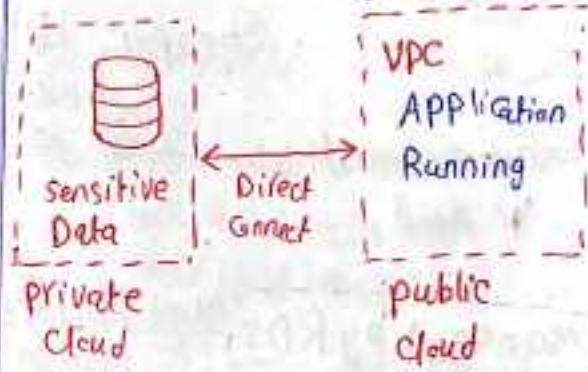
Health checks on AWS Resources
 ← تقدر كمان
 عشان تو ديك لا عوزي نقطة شغاله ولو ال Server واقع مش هيد ال route عليه

API Gateways
 ← allow to share and manage data between systems by creating APIs
 ← Integrated with other services like Lambda



اسهل
 ← و لكن سرعة النقل ال data ابطا

Super Fast Transfer
 ← ال transfer هيا هياكون
 و النقل بيتم على ال [AWS private Network] vpc
 برودو زي ال Peering



use Cases
 1. Transfer Large Datasets
 2. Transfer Business Critical Data

In Hybrid Model
 2. **site-to-site VPN**
 ← ودا بيكون ال Connection through Internet

← ال Data هياكون اتشفرت
 ← ال option دا ال automatically encrypted
 ← Direct Connect

← تابع ال replica وال replica عبارة

عن [Read-only Copy] of DB For Fast querying
ولما بعد replica على أكثر

من Region دا لحد حسن ال Performance
ولم يزد ال durability.

2. Relational DB ← Aurora

Compatible with

MySQL (x5 Faster normal)

PostgreSQL (x3 Faster normal)

⊕ انما cheaper

← دا غير انما scale automatically in need

← Aurora هو Managed by RDS

3. Fully managed ← DynamoDB

NoSQL DB
(Key-Value DB)

Serverless

لاحظ انه ال RDS
مكانيش
Serverless
لانك بتصدر ال instance type
وهذا

Fast Performance [milliseconds]

scales automatically

بردو هي
كان

← Data Bases * organized

Collection of various forms of data, that used by many applications.

1. Relational Database ← RDS Service

service that launches & manage relational DBs, it supports popular DB engines like: Aurora, PostgreSQL, MySQL, MariaDB, Oracle, SQL Server

← عذو Fault ⊕ high availability tolerance

لدي عذو deployment option انه يكون على Multi-AZ

← AWS هو ال سيقلا Software Patching, automated backups

os maintenance, ... ← عذو ميزة مهمة حد اوهي

انك تقدر [Launch read replicas across Regions]

Real World use Cases For all Database Services:

1. Migrate on-premises Oracle DB to Cloud → RDS

2. Migrate on-premises PostgreSQL DB to Cloud → Aurora or RDS

3. Alleviate DB load for data that is accessed often → ElastiCache

4. Process large sets of user profiles and social interactions → Neptune

5. NoSQL database fast enough to handle millions of requests per second → DynamoDB

6. Operate MongoDB Workloads at scale → Document DB

Migration and Transfer

Database Migration Service (DMS)

help to migrate database to AWS

← migration دا ممكن يبقى نوعين

نوع اول (Oracle to Oracle) وفي الحالة دي

الاسم homogeneous او نوعين

ممكن زي بعين وهما اسم heterogeneous

وال DMS بتقدر تقول النوعين عادي.

Document DB: 4

Fully managed document DB that supports MongoDB

داتوع تاني لا تالو NoSQL مختلف عن ال Key-Value

Serverless Fully managed

ElastiCache: 5

Fully managed in-memory data store

that compatible with Redis & Memcached

طريقتين ال in-memory data stores

are used to cache and accelerate data retrieval in applications

← (لا حفظ الـ تخزين جوا ال memory للـ شئيه بين ال [Data Can be lost])

offers high Performance + low latency

Neptune: 6

Fully managed graph database that supports highly connected data sets.

Like Social media networks

server less Fully managed

← كمان ال Fast + Reliable

AWS Snow Family 3

لأنه يستخدم مهم ما يكون عند large amount of data
ولو نقلنا على النت ممكن النقل يأخذ أكثر من اسبوع
Snow Family allow you to transfer data from on-premises to the cloud use physical device, والنقل سيكون through الإنترنت

لأن في الحالة نقل على الإنترنت يكون
عشان البنية التحتية with high cost

Snow Cone 1

هو the smallest member

وبد Terabyte 8 فيتر hold بقدر

يعتبر using offline shipping أو online with

DataSync

Snowball & Snowball Edge 2

لأن سعة في النقل Peta byte

وبد transfer data في الابقاهين in out
(يعني ممكن من cloud إلى on-premise)

مع داعية ان هو في حدة ااته Computing device that support EC2 & Lambda

بمعنى انك لو حلال في مكان معين شانت وعاوز
على ال processing تستمر هو هيعمل كدا
في وهنا هو cheaper than internet transfer

س تايه DMS

لأن migration دا بيتعم عن طريق
Continuous data replication
و يكون No downtime (يعني)
ال database بتعقد تنقله وهما بتتقد
عادي جدا

Real World Use Cases

1. Oracle on-premises to Aurora
MySQL on Cloud

2. Oracle on-premises to
Oracle on EC2 (on Cloud)

3. RDS Oracle to Aurora MySQL
(Cloud migration من cloud)
(نقل عادي)

Server Migration Service (SMS) 2

allow to migrate servers from on-premises to Cloud (group of servers)
(كمان ميتا واحد)

لأن ال Servers ال هنتقل على بيتم لها
Save as Amazon Machine Image (AMI)
وبعد ها كدا

ناخذ ال AMIs دول نروح ز
Launch EC2 Instances with them
و بعدا تكون ال Servers اتقلت

data Replication لا
→ Cross-Region or Cross-account

Datawarehouse

← Analytics *

→ is a data storage solution that aggregate a massive amount of data from disparate sources

← مستخدم داخلي في اثنائها Querying, Reporting, analytics - - - -

← في فرق مهم جدا انها انه ال data للتخزين في DBs زي RDS والار اكسنا عليهم من شوية دول بيستخدمو في ال online transaction processing
[update, insert, delete - - - - Transactions كثير جدا زي ال delete - - - -]

← اما ال data warehousing مو ل transaction processing

لا دول بشكل mainly في ال [analytics + reporting]

It's a scalable datawarehouse solution that can handle exabyte-scale data

when you use it, it improves speed and efficiency when querying.

Redshift Real world use cases

1. Data Consolidation
consolidate multiple data sources for reporting

2. Relational Databases
you want to run a DB doesn't require a real-time transaction processing (insert, update, delete)

snowmobile

Multi-Petabyte or exabyte

← سفته في النقل
غالبا بتستخدم ال option
دا لما بتبقى

completely shutting down your on premises DB and moving totally to the cloud

← ال Data هتقل هنا في S3

← على الرغم من ان النقل في عربية ولكن

[securely transported]

فندق GPS tracking alarm monitoring

24/7 Video Surveillance
← مراقبة الفيديو

4. Data Sync
allow to transfer data online from on-premises to the storage service in the cloud like S3 or EFS - - - - with data transfer speeds are up to 10 times faster than open-source tools.

← تقدر ت Copy ال data
أو over the Internet
with Direct Connect

← متى بيس هتقل migration من ال on premises
لا انت كمان تقدر تقول بيا (cloud)

[copy data between AWS storage services]

Elastic map Reduce ← EMR .5

is a managed big data platform that includes popular ETL tools like Apache spark and Apache Hadoop.

من الآخرة هي لتعمل big data processing
عشان أو كمان نقدر نعمل بها data mining
machine learning وأي حاجة في الدنيا بتعمل

[large amount of data] مع

service that ← Data Pipeline .6

automate movement and data transformation between various services
⊕ on-premises و كمان data sources

يعني من الآخر هتعمل orchestration
for data automate movement between services

فهتعمل بقي في حركة و نقد البيانات في فترات محددة

[moves data at specific intervals] on-prem sources

[move data based on conditions] كمان هتعمل

[send notifications on success or failure] كمان

: analytics ←

is the act of querying or processing the data.

is a query service ← Athena .2
for s3 (analyze data in it using SQL)

في ال service دي أنت pay per query

serverless

Prepare the data ← Glue .3

For analytics (مستعملو Cleaning وكذا) [better understand] هتخلي عندك

يعني بتعمل ETL service

processes (Extract, Transform, Load)

على ال data في ال Warehouse

analyze data ← Kinesis .4

in real-time in Videos and Streams.

بيقول كمان انه real-time عامة في أي حاجة زي

Video ⊕ audio

application logs (البتصل حلا) (live data)

websites clickstreams

IOT وحتى كمان ال

is natural-language **Comprehend** .2
Processing (NLP)

Service that find insights and relationships to **analyze text**.

Comprehend in Realworld **التفهم في العالم الحقيقي**

1. Review social media posts

Comprehend looking in social media posts for [words]

ويعمل تجزئة الكلام و يفهم العلاقة
المابين الكلمات و بالتالي يفهم رأي
الشخص الكاتب داو معناه بلفظه بالطلب

يعني من الآخر يفهم اللغات البشرية
natural languages

Turn text to speech **Polly** .3

كمان ببعده كدا وهو بيقدر صوت بشري
[mimics natural-sounding human speech]

وهو عنده default

[several voices across many languages]

Create Custom Voice
ومش بس كدا انت كمان تقدر
Voice

لا حظ علامته speech

ال بيطع output كدا انت تقدر تحفظ به
mp3 و تستخدمه عادي.
audio

help you to **QuickSight** .7

Visualize your **data** ⊕

Build interactive dashboard,

you can also embed this dashboard to your app.

Analytics Real World Use Cases

1. search in S3

Athena

[query historical data in S3 using SQL]

2. Log analytics
Kinesis

[analyze in real-time for app monitoring ⊕
Fraud detection]

Machine Learning #

analyze images & videos and

Rekognition .1

[identify Custom Labels]

زي انه يعرف يحدد ال objects ال في الصورة
و يستخرج text و يعرف على الوجه
كمان

[Face ⊕ Text Detection in images and videos]

يعمل object detection من الآخر.

Developer Tools

Integrated Development Environment (IDE) within your browser to write and debug your code. ← هو عبارة عن

Cloud 9 in Real world use case

Build a serverless app

preConfigure Cloud 9 IDE development environment

Code in your browser

local IDE

source control system for

Git repositories (Github)

create repositories + commit, branch, merge the code

collaborate with other developers

Build + Test your app source code

Compiler + run test cases

enable CI/CD → artifact deployment

help you to build, train and deploy ML models. ← SageMaker

وبعد دا عن طريق

prepare data for models

Train & deploy the model

provides Deep learning

AMIs

EC2 instances with

high compute capacity that accelerates machine learning and deep learning in the cloud.

SageMaker in real world

Make Recommendation engine (by ML models)

Provides real-time + batch language translation

يكون استخدمه حلوشا

Websites Localization

help to Build conversational interfaces like chatbots

understand + recognize speech languages

integrate voice into device

debug production app
⊕ analyze it

[app Component mapping] بتقيل
Frontend app الخونة
[view Request send to end] كمان

X-Ray 6

التوضيح

X-Ray in Real World

[X-Ray can map requests that made to RDS from your app,

فقط رقائق information عن اى ال queries
الحصلت وراحت حايبة اى وعلت اى

(يعني حرفياً بت trace و debug و analyze
كل حاجة بين ال Components المكونة
لا بلاتش)

Code Build Real/World Use Case

Run tests before deploy
an new version of production
system.

manage the deployment
of code on [compute services or on-prem] 4

maintains app uptime (keep it available)
دا عيز انهارب
يعني

[Rolling out a new version of app]
حتى لو انت بت

Code Pipeline 7

used in CI/CD processes

integration و بيحلو

مع برود Code Build و Code Commit و Code Deploy

بس ال purpose هو باختلاف دي بتركز انهارب
Simplify development process

[Work Collaborating on development projects] developer's
Contains issue tracking dashboard
دا عيز انهارب

manage development pipeline

Code Pipeline 5

automate software release processes

[quickly deliver new features & updates]

Integrate with Code Build
to run builds & unit tests

Integrate with Code Commit to receive source code.

Integrate with Code Deploy to deploy your changes

Code Pipeline in Real World

automate build, test and deployment of your app.

apply develops practices that automate the movement of code to production

Elastic Beanstalk

allow to deploy web apps and web servers to AWS

automatically handle deployment from Capacity Provisioning, load balancing and auto scaling.

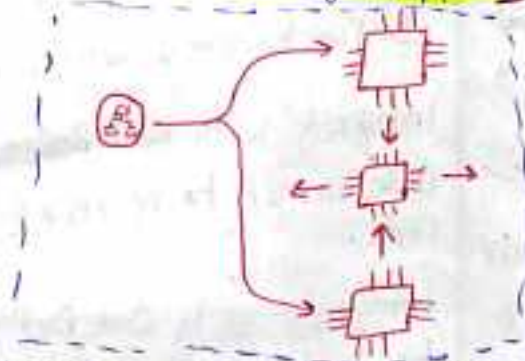
also monitor application health via health dashboard

Elastic Beanstalk in Real World

Quickly deploy Java-based web app to AWS

after upload Java code Elastic Beanstalk handles Capacity Provisioning, load balancing and auto scaling.

Monitor the health of your app



Automatic resource generation

الكود يتابعك Elastic Beanstalk

Compute Service

behind the scene

Deployment & Infrastructure Management

Infrastructure as Code (IaC)

allow you to write scripts to provision resources, its benefit is in reproducible manner several times

We write these scripts with json or YAML

Provision AWS Resources via scripted template (IaC)

CloudFormation

CloudFormation in Real World

use it to automate creation of EC2 instances in AWS account

2. Elastic Beanstalk

It designed for developers who want to write code and not manage infrastructure

CloudFormation is IaC

Infrastructure building

Elastic Beanstalk

Simple interface

mainly on Deployment

create infrastructure automatically

cloud formation

messages are processed in asynchronous manner

→ [process ^{يعمل بالتوازي}
 → [wait ^{ينتظر} للتانية ^{لحظة}
 → [ما تخلص ^{فما تخلص}

SQS in Real World

→ Build money transfer app that performs well under heavy load

[Requests processed in **FIFO** order ^{نظام}
 Queue ^{لأنه يتلقى الطلبات وينتظر}

Simple Notification Service

SNS

2

→ [plain text emails]

Send emails & text messages from your app and make publish to a topic

→ subscribers ^{الذين يتلقون الرسائل}
 receive messages

SNS in the real World

→ Send an email when CPU utilization above 80%



work with **ops works**

chef and **Puppet** to automate the configuration of your servers using code.

[on-prem ^{دواء}
 EC2 ^{أو على ال Cloud}
 ^{في بيئتك على الإنترنت على}

ops works for Real World

→ ops work allow you to define software installation scripts and automate configurations for servers

"Automate infrastructure configuration & management for your app"

Messaging and Integration

Loose coupling helps reduce the risk of cascading failures between components.

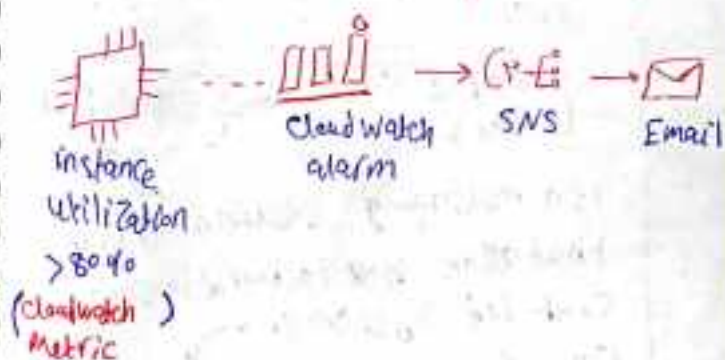
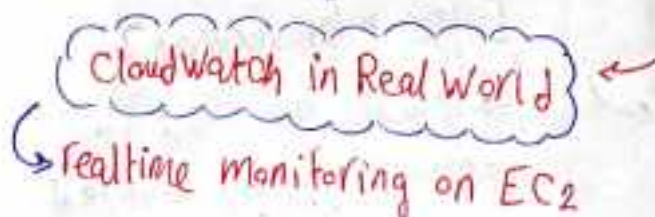
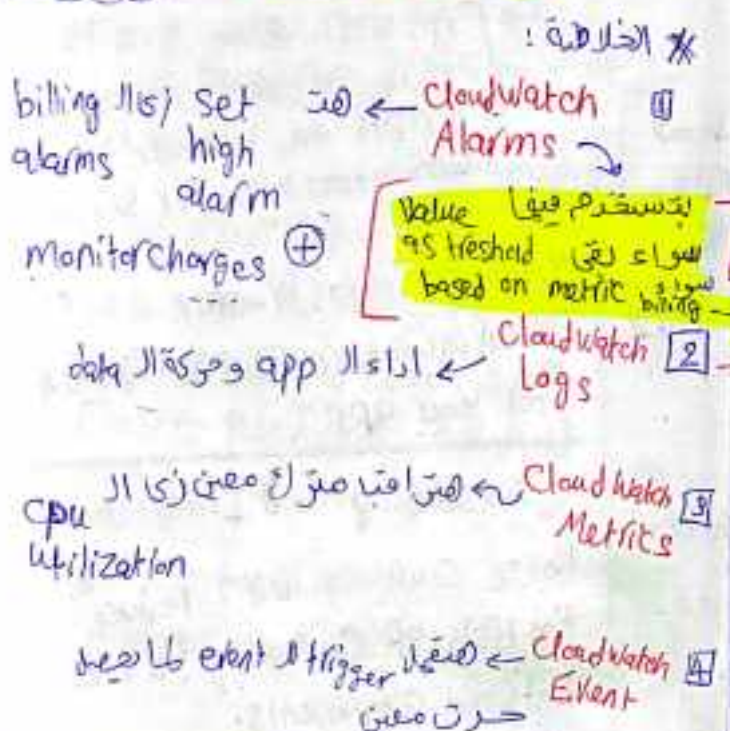
[use microservices]

→ Queues are used to implement loosely coupled systems

[simple Queue Service] **SQS**

is a messaging queuing service that allow you to build loosely coupled systems.

[allow component to component communication via message]



allow to send richly formatted HTML emails from your app

It is ideal choice for marketing campaigns or professional emails.

SES in the Real world

→ send marketing email and track open or-click through rates

Auditing & Monitoring and Logging #

cloudwatch [1]

Collection of services that help you monitor and observe your cloud resources

Collects metrics, logs and events

detect anomalies in your environment

set alarms ⊕ Visualize logs

CloudWatch Services

CloudWatch Alarms [1]

set high resolution alarms

billing alarms

CloudTrail in RealWorld

→ you can troubleshoot events over the past 90 days using CloudTrail event history log to find a specific time event per-region [تقدر Customize المدة التي تم من 90 يوم عادي]

Things you can track with CloudTrail

1. Username

2. Event time and name

3. IP address

4. Access key (track access key used)

5. Region

6. Error Code

Additional Services *

1. Workspaces: host a virtual desktop in the cloud

[Windows or Linux Desktops]

That enable employees to work from home.

2. Amazon Connect: Build a cloud contact center or helpdesk in the cloud

[provides customer services functionality + improves productivity of help desk agents]

another example Cloudwatch In Real World

→ Receive notification when root user activity is detected in your account

[root user activity is detected in your account] من خلال root user API Calls

وبالتالي هنقدر نقدر Trigger rule [with Cloud watch Event rule]



2. Cloud Trail: Track user activity + API Calls

Through Console, SDK, CLI...

→ Tracking and Identify which user made changes + Detect unusual activity in the account

Log and retain account activity (تسجيل والحفاظ على نشاط الحساب)